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*Issue: Resources and Technological Advances for Studies of Neurobehavioral Evolution***Brain collections at the National Museum of Health and Medicine**Archibald J. Fobbs, Jr.¹ and John I. Johnson²¹National Museum of Health and Medicine, Armed Forces Institute of Pathology, Washington, DC. ²Radiology Department, Division of Anatomy, and Neuroscience Program, Michigan State University, East Lansing, Michigan

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Owing in large part to the foresight and efforts of Wally Welker, the National Museum of Health and Medicine has become a major repository for collections of brain specimens vital to the study of neurobehavioral evolution. From its origins in the Armed Forces Institute of Pathology, with the collection of largely pathological specimens assembled by Paul Yakovlev, the museum has added to its resources four additional extensive collections, largely consisting of specimens acquired specifically for comparative and evolutionary studies: Welker's collection from the University of Wisconsin-Madison, John I. Johnson's collection from Michigan State University, the Adolf Meyer Collection from the Johns Hopkins University, and the Elizabeth Crosby collections from the University of Michigan. We describe here the history and contents of each of these five collections, to inform the scientific field of the extent and details of these remarkable resources.

Keywords: neuroanatomy; evolution of brains; comparative neuroscience; neurobiodiversity; brain databases

Introduction

Wally Welker, as he was nearing retirement, searched for a home for the extensive collection of brains in the Physiology Department at the University of Wisconsin, including his own acquisitions, those assembled by Clinton Woolsey, and some specimens brought by Jerzy Rose from the Meyer Phipps Collection (see below) at Johns Hopkins University, where this valuable material would be protected from destruction consequent upon administrative changes in missions and goals in university structure.

His quest found a like-minded response from Adrienne Noe, in charge of collections at the National Museum of Health and Medicine (NMHM)

at the Armed Forces Institute of Pathology (AFIP), located in the Walter Reed Army Medical Center in Washington, DC. The NMHM was the major repository of collections of embryos from major sources around the country and also possessed the great Yakovlev-Haleem brain collection (see below). Noe and Welker saw the addition of the Wisconsin collection to the Yakovlev-Haleem as a first step in forming a similar central national home for brain collections.

Welker secured significant assistance in identifying sources of support, from Christopher Platt, then at NSF, who also saw the great advantage of bringing collections together at a single location. This consolidation greatly facilitates broad studies by researchers, and aids teachers seeking instructional materials, in a sanctuary protected from the vagaries of scientific ebbs and flows in current topical materials. Scattered university facilities are necessarily at risk, and many specimens and collections have been lost to these winds of change. This joint collection principle attracted more additions to the critical mass: brains from Michigan State University (see

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below)—an adjunct of the Wisconsin collection—the great and historic Crosby Collections (see below) at the University of Michigan (not to be confused with Michigan State University), from the Meyer Phipps Collection—the ancestor of UW and MSU collections—and other smaller groupings, including the Pubols Collection from Oregon Health Sciences University, the Lockard Collection from the Medical University of South Carolina, and the Starr Collection from Columbia University.

Financial support for the transfer of the Wisconsin, Michigan State, and Michigan collections was provided by the National Science Foundation, along with other sources that assisted in the general process of collection amalgamation. The value of the brain and embryo collections was recognized in the federal law governing the closure of the Walter Reed Campus in assuring the protection and continuation of the NMHM as a national resource.

In the tables of contents of each of the collections presented below, we have preserved, as far as possible, the format and nomenclature used by the original curators of the collections. To alter and make assumptions about what is meant by entries in the catalogs risks distortions and errors; users of the collections will be better able to judge what contemporary nomenclatures and classifications will be appropriate for their purposes.

The Yakovlev-Haleem Collection

History of the collection

Paul Yakovlev was born in 1894 and received his degree in medicine from the Military Medical Academy in Leningrad in 1919. He left Russia later that year, walking across the frozen Gulf of Finland with only a satchel and his medical diploma.¹

Eventually, he migrated to Paris where he studied under the famous neurologists Pierre Marie and E. J. Babinski. He received an M.D. degree from the University of Paris in 1925. In the same year, he emigrated to the United States and practiced medicine in Providence, Rhode Island. He met Stanley Cobb and William G. Lennox of Harvard University and worked under them in the neurology clinic of Boston City Hospital. He became the liaison between Harvard and the Monson State Hospital at Palmer, Massachusetts. According to Yakovlev, “that’s where I began my career as a scholar.”¹ Lennox had wanted him “to make some sense” out

of the epileptic patients at the hospital and their disorder. In contrast to his vocation of neurologist to the patients, “by avocation” as Yakovlev put it, he began to study by autopsy some of the brains of his former patients.

Lacking proper facilities and techniques for proper preparation of autopsy material, Yakovlev secured a fellowship to study in Switzerland at Constantin von Monakow’s Hirnanatomisches Institut in Zurich, where he studied with M. A. Minkowski. At the institute he vividly recalled long lines of cabinets containing serial sections with “some millions of slides.” Upon return to Monson he looked into the construction of a proper microtome to adequately prepare sectioned brains for study, as well as staining and mounting techniques, and thus was born the eventual great Yakovlev Collection of normal and pathological human brains. He continued this enterprise in his later assignments.

In 1936, Yakovlev became director of laboratories at the Metropolitan State Hospital in Waltham, Massachusetts. In 1938, he became clinical director of research laboratories at the Walter E. Fernald School in Waltham, Massachusetts, where he prepared many of the specimens in this famous collection of serial histological sections of over 1,000 brains. All the slides (over 250,000) in this collection were carefully prepared, sectioned and stained by the same exacting technique. The collection moved to the Warren Museum of Harvard University where Yakovlev became associate clinical professor of neuropathology and curator of the Warren Anatomical Museum.

After his retirement, the collection moved back to the Fernald School. In 1974, by a joint agreement with Yakovlev, Harvard University, the National Institutes of Health and the AFIP, this outstanding collection was transferred to the AFIP and its NMHM in Washington, DC, where it is now available as a national resource for the research studies by qualified scientists.

In addition to the specimens acquired personally by Yakovlev, many specimens were donated to the collection from a great variety of sources. For comparative and evolutionary research, notable contributions include monkey brains from Mortimer Mishkin, of the National Institute of Mental Health, and from Samuel Gelfan and others at the Yale Aeromedical Research Unit at Yale University; as well as dolphin brains donated by Peter

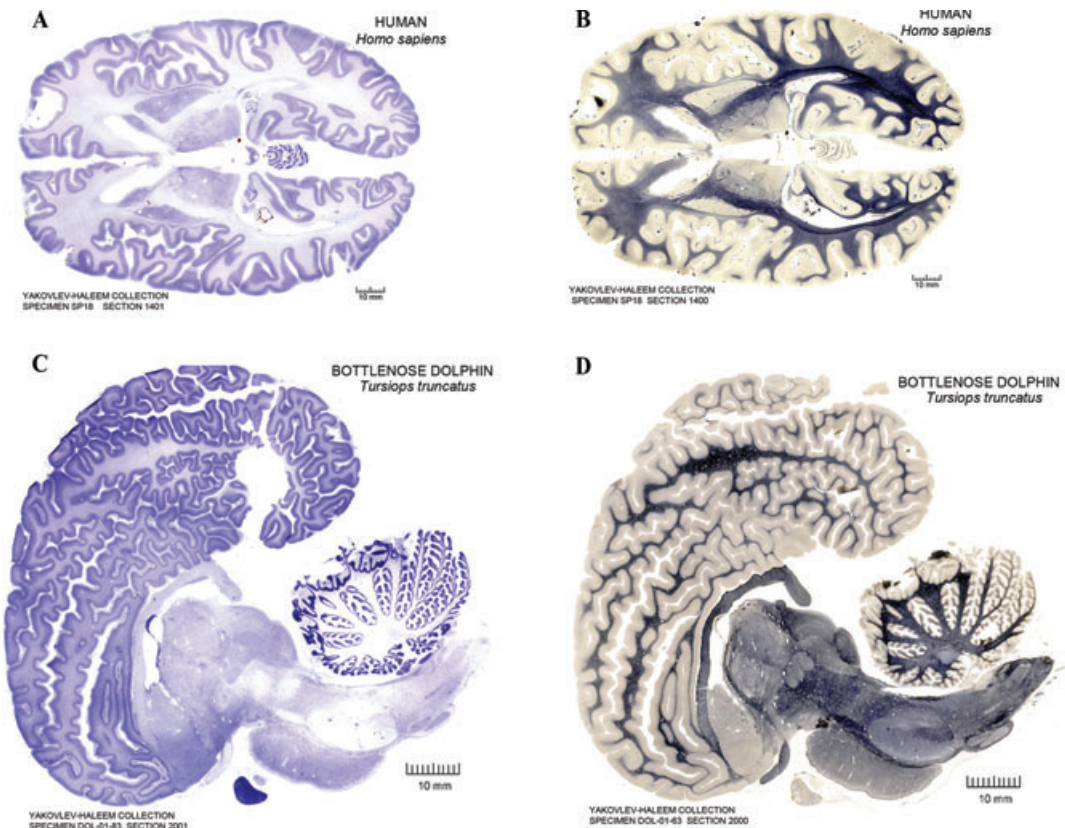


Figure 1. Illustrations of stained sections in the Yakovlev-Haleem Collection. Images A and B show sections in the horizontal plane, C and D in the parasagittal plane. Stains used with these sections are cresyl violet, Nissl method in A and C, and iron hematoxylin, Loyez version of the Weigert method in B and D. Fully labeled copies of these images can be seen in online atlases at <http://www.brains.rad.msu.edu>, or <https://www.msu.edu/~brains/brains/index.html>.

Morgane from the Communications Research Institute in Miami, Florida.

Mohammed Haleem was associated with Yakovlev's brain collection for more than thirty years. He was born in Palestine, and came to the United States in 1960, settling in Boston, Massachusetts where he first began working with Yakovlev. Haleem came to the AFIP in 1974 when Yakovlev relocated the collection to AFIP from Harvard Medical School. In 1983, Haleem became curator of the Yakovlev Collection, which, in honor of his dedication and contributions, was renamed the Yakovlev-Haleem Collection in February 1994. Haleem is remembered today for his unmatched skills in preparing brain slices for study.

Contents of the collection

Most of the brains in the collection were embedded in celloidin, and were sectioned at intervals of

35 μ m. The stained sections are mounted in pairs of stained sections at intervals of from 5 to 25 sections. One member of each pair is stained with cresyl violet by the Nissl method to show cell bodies; the other is stained with iron hematoxylin by the Loyez modification of the Weigert method to show myelinated fibers (Fig. 1). Many of the intervening, unstained, sections are preserved in individual containers for possible future use.

The collection is catalogued here in two parts. The Human Collection (Table S1A) includes 433 human brains specimens designated as "normative controls" in what constitutes a developmental series, ranging in age from embryos to the 10th decade. The Comparative Collection (Table S1B) consists of brains of three cats, 3 embryonic fur seals, 20 embryonic and adult whales and dolphins, 1 horse, 38 chimpanzees, 1 gorilla, and 97 fetal and adult rhesus monkeys.

In addition to the specimens listed here, the Yakovlev-Haleem assembly also includes another 576 specimens of a great variety of pathological conditions, including 129 malformations, 149 instances of miscellaneous neuropathology, 49 cases of cerebrovascular strokes, and 92 results of neurosurgery, as classified by Kretschmann *et al.*²

The Welker Wisconsin Collection

History of the collection

This history is based on that compiled by Wally Welker and published on the website <http://brai nmuseum.org>.

The Welker Collection was started in the 1950s by Clinton Woolsey. Woolsey came from Johns Hopkins to start brain research at the University of Wisconsin–Madison, soon after the Second World War. He was the founder of the laboratory, and later the Department of Neurophysiology at Wisconsin. He brought with him not only a strong and extensive interest in the comparative anatomy of mammalian brains, but also the beginnings of what has become one of the premier collections of its kind anywhere in the world.

Konrad Akert played a role in standardizing techniques of perfusion and histological processing and protocol keeping in the early years. The brain collection was increased and diversified by Wally Welker and colleagues over the next four decades. During this period, a large number of faculty and students carried out comparative neurophysiological and neuroanatomical analyses in a variety of mammals.

Many of these studies were initiated because of the special behavioral repertory of the different animals studied. John I. Johnson joined these investigators for postdoctoral training in 1960. By the time Johnson arrived, Welker, a member of the Wisconsin faculty, had begun the formal collection of brain specimens with five-year support from the National Institutes of Health for “comparative study of brain morphology in mammals.”

With the advice of Konrad Akert and Jerzy Rose at Wisconsin, together with the collaboration of numerous students and colleagues, Welker assembled brains from a wide variety of mammals from a representative variety of mammalian orders—specimens that were perfused and prepared by standardized histological procedures in ways that would provide

neuroanatomical study materials for many generations of comparative neuroanatomists.

Welker worked together on these comparative projects with Johnson over the last several decades and both worked together with Jon Kaas, John Allman, Michael Merzenich, and many others over this time span. The neuroanatomical material that was assembled was transported and is now curated and maintained at the NMHM in Washington, DC, under the supervision of the museum’s director, Adrienne Noe, and curator, Archibald Fobbs, and is made widely available for study by students, educators, and researchers worldwide.

Contents of the collection

The Welker Wisconsin Collection consists in a total of 409 specimens with representatives of 122 species. Most of these are mammalian brains. A few specimens of brains from other vertebrate clades are included: amphibian salamander and bullfrogs; reptilian alligators, Tegu lizards, iguana, Western rattlesnake and turtles; and avian ostrich, owls, gulls, and pigeons (Table S2). Some special features are a developmental series of brains of rats and cats and an extensive experimental series of brains of echidnas, galagoes, squirrel monkeys, chimpanzees, beavers, mice, rats, rabbits, cats, raccoons, and manatees.

As collecting proceeded through the years, a standard protocol was gradually established, including metric photography of the living animal as well as of fixed specimens of bodies and brains, and the manufacture of acrylic casts of the brain of one specimen from each species.

The brains were then embedded in celloidin, or in some special cases in paraffin, and were sectioned at intervals of 15–60 μm , depending on the size of the brain. The stained sections are mounted in pairs of stained sections at intervals of from 1 to 10 or more sections, again depending on brain size. One member of each pair is stained with thionin by the Nissl method to show cell bodies; the other is stained with iron hematoxylin by the Weil, Mahon, or Sanides-Heidenhain-Woelcke³ modifications of the Weigert method to show myelinated fibers (Fig. 2). The sectioning and staining, as seen in this figure, was done in accordance with very exacting standards by a team of long-working histological technicians, supervised for many decades by Helen Brandemuehl, who had been personally trained by Cecilia Bisson, Adolf Meyer’s chief

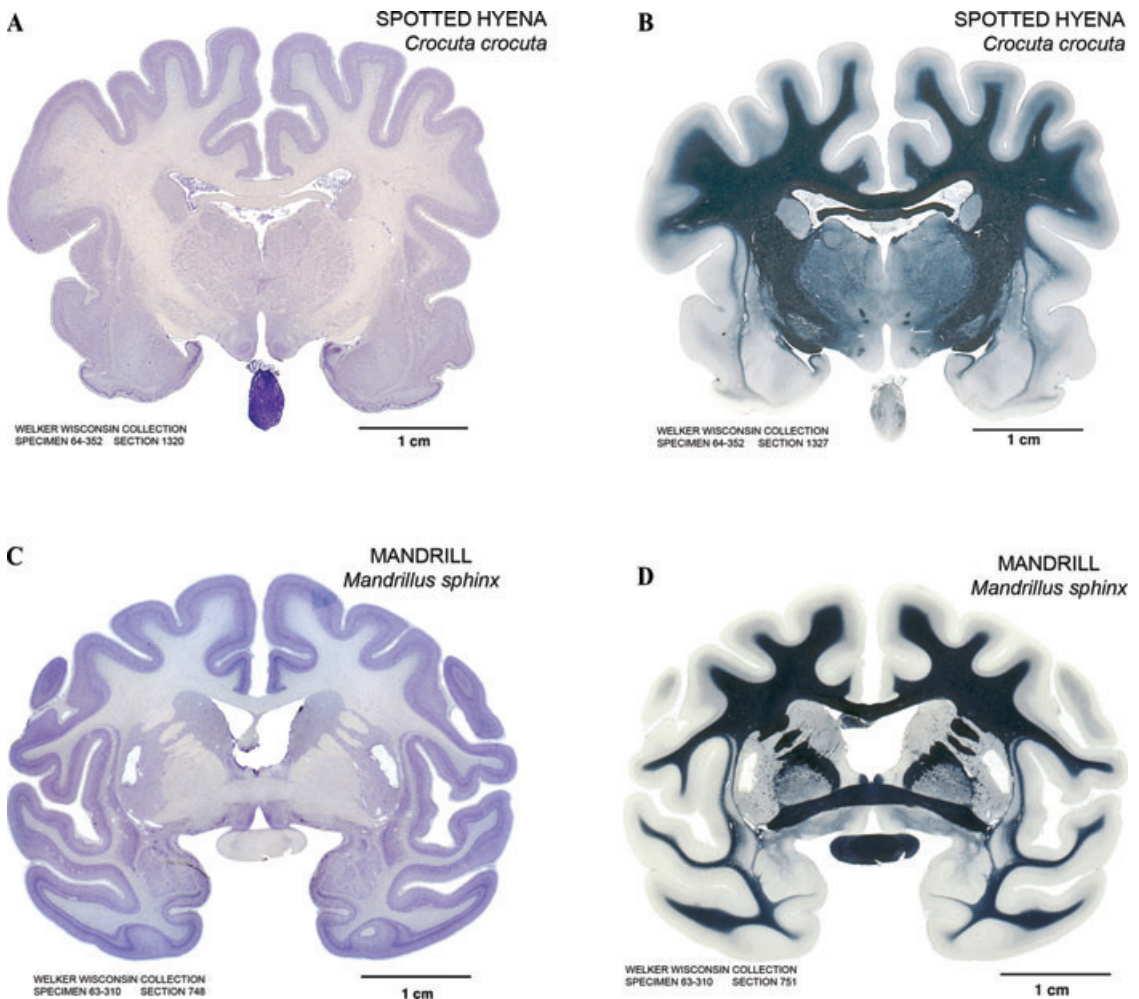


Figure 2. Illustrations of stained sections in the Welker Wisconsin Collection. Sections are in the coronal plane. Images A and C are stained with thionin, Nissl method; B and D are stained with iron hematoxylin, the Sanides-Heidenhain-Woelcke modification of the Weigert method. Additional illustrations of specimens in this collection can be viewed at <http://brainmuseum.org>.

histologist at the Hopkins Phipps Clinic as arranged by Clinton Woolsey when he founded the laboratory; in later years the Bisson-Brandemuehl tradition was continued by Inge Siggelkow.

Johnson Michigan State Collection

History and contents of the collection

The arrival in the Wisconsin laboratory of a zoo specimen of an Australian Western Gray Kangaroo, in 1962, highlighted a notable gap in the Wisconsin collection. There was a dearth of specimens from the great marsupial branch of the mammalian radiation. John I. Johnson was stimulated by this discovery to plan an expedition to Australia to fill

this gap, by means of a Fulbright fellowship and grant support from the National Institute of Mental Health. With much assistance from Peter Bishop, his Australian host, at that time professor and chair of physiology at the University of Sydney, and Michael Marsh in the Sydney Zoology Department, he was able to obtain specimens of many local marsupial species. In addition brains were obtained from species of Catshark and avian Emu from experiments in cardiovascular physiology in the Sydney department.

Marsh introduced Johnson to John A. W. Kirsch, a graduate student at the University of Western Australia who was in Sydney to obtain specimens of

various tissues from eastern Australian species for his dissertation research, with W. D. L. Ride at the University of Western Australia, on marsupial phylogeny as revealed in comparative serology. Through this connection, and the assistance of Peter Bishop, he was able to travel to Perth, where with Kirsch's assistance specimens were obtained from the colonies maintained in the UWA Zoology department. Later Kirsch and Johnson jointly toured Tasmania, with excellent assistance from the Animals and Birds protection board of Tasmania, and were able to obtain from their native habitats specimens of Tasmanian Devil (now in danger of imminent extinction), Native Cat, Tasmanian Pademelon, Rufous Wallaby, Potoroo, and a monotreme *Echidna*.

The Johnson and Kirsch collaboration continued during Kirsch's postdoctoral fellowship at the University of Kansas, which included an expedition through South America gathering specimens of the South American marsupial radiation. While obtaining his various tissue and serological specimens for further work in marsupial phylogeny, Kirsch, after his experience in Australia, was able to also perfuse brains for addition to the brain collections, including such phylogenetically significant and rare specimens as *Caenolestes obscurus*, *Lestoros inca*, and *Dromiciops australis*. Additional South American species, including marsupials *Didelphis albiventris*, *Marmosa murina* and *Thylamys velutinus*, were kindly donated by Eduardo Oswaldo-Cruz and Carlos Rocha-Miranda at the Federal University of Rio de Janeiro, as were specimens of placental agoutis by Drs. Angelo Machado and Gilberto Campos at The Federal University of Minas Gerais in Belo Horizonte.

Another significant gift to the collections was a complete set of "extra" unstained sections of brains of several species of bats, mostly Caribbean, by Dr. Paul Pirlot of the Université de Montréal. These sections, embedded in paraffin, were subsequently mounted and stained at Michigan State. They are likely to be the only survivors of the large Pirlot collection; the rest of the collection was destroyed in a disastrous fire in Quebec.

The Michigan State collection also includes normal specimens of several species maintained at the university for experimental research, including *Anolis* lizards, bobwhite quail, sheep from the endocrine research unit in physiology, kangaroo rats, various species of *Peromyscus* deer mice from var-

ious habitats from the colony of John A. King in zoology, as well as local opossums, fox squirrels, and raccoons from experiments in the Johnson laboratory.

A complete listing of specimens in the collection includes 81 species, represented by 177 specimens, 83 placentals, 13 bats, 69 marsupials, 5 monotremes, and 19 non-mammals (Table S3).

Processing of all these specimens followed, as far as was possible, the protocols and procedures used at Wisconsin by Wally Welker, including photography of the living animal as well as metric photographs of fixed specimens of bodies and brains; manufacture of acrylic casts of the brain of one specimen from each species, embedding in celloidin and staining with alternate sections series stained with thionin by the Nissl method, and with hematoxylin to show myelinated fibers, using either the Weil or the Sanides-Heidenhain-Woelcke³ variants of the Weigert technique (Fig. 3). Sectioning, staining, and mounting of the stained sections on glass slides was the work of many skilled technicians and students over the decades, with notable contributions by Grace Heung-Sook Chung, Tony Duke, Sanford Feldman, Emmeline Gatewood Haight, John R. Haight, Paul Herron, Joanna M. Hill, Michael Ostapoff, Michael Peterson, Edwin Rubel, Sharleen Sakai, Steven Warach, Michael Widener, Robert C. Switzer III, Philip Ulinski, W. Lee Weller, and Sidney Wiener.

Meyer Phipps Hopkins Collection

History of the collection

In 1908 the Philadelphia steel magnate Henry Phipps visited the tuberculosis clinic he had funded at the Johns Hopkins University, and asked if there were another opportunity for his philanthropy. Dean William Henry Welch of the Hopkins Medical Faculty took the opportunity to found a department of psychiatry, in a state-of-the-art clinic funded by Phipps, and hired Adolf Meyer to oversee the foundation of the clinic, department, and curriculum. Meyer was Swiss, and professor of psychiatry at Cornell University at that time; he believed in a scientific basis for Psychiatry, emphasizing biological factors and life experiences as explanatory, preferring data-based to theoretical formulations. He originated the terms and the fields of biopsychology, and of mental hygiene. This thinking impressed Dean Welch, and this was his basis for recruiting

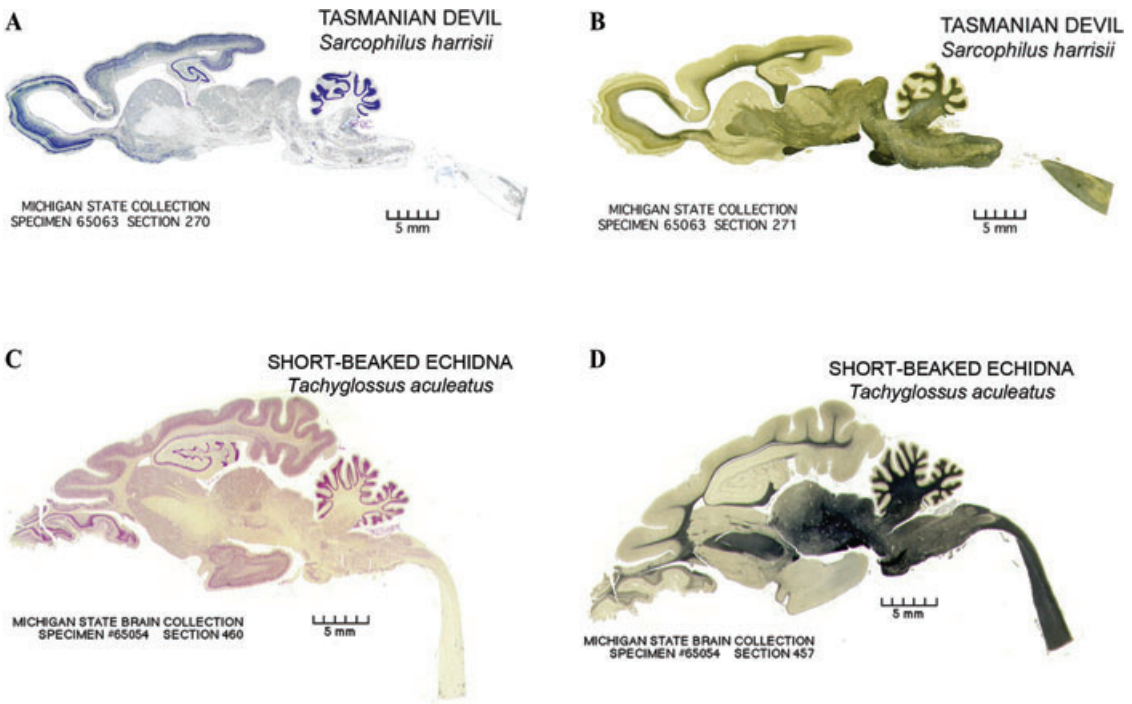


Figure 3. Illustrations of stained sections in the Johnson Michigan State Collection. Images A and C are stained with thionin, Nissl method; B and D are stained with iron hematoxylin, B using the Weil modification of the Weigert method, and D the Sanides-Heidenhain-Woelcke modification. Additional illustrations of specimens in this collection can be viewed at <http://www.brains.rad.msu.edu> or <https://www.msu.edu/~brains/brains/index.html>. Under Title 17 of the U. S. Code, Section 105, copyright protection is not available for any work of the United States Government. The opinions or the assertions herein are the private views of the authors and are not to be construed as official or as reflecting the views of the National Museum of Health and Medicine, the Department of the Army, or the Department of Defense.

Meyer to oversee the development of psychiatry at Hopkins.⁴

Accordingly the clinic at Hopkins would have patients, laboratories, and experimental and observational research, including extensive case histories and anatomical and physiological laboratories. Meyer had, in his own training in Switzerland, studied with the renowned neuroanatomists Auguste Forel and Constantin von Monakow (at the same institute where Paul Yakovlev learned about brain collecting in subsequent years). As a result, Meyer, like Yakovlev after him, believed in having adequate anatomical and physiological tools for psychiatric research. This included an ever-growing collection of neuroanatomical material, including brains of animals used in experimental research.

The neuroanatomical material was prepared using the best technology and standards, and in Meyer's long tenure this resulted in the extensive brain collection described here. Many classical

anatomical studies resulted from use of these materials beyond the reasons for their initial collection. For example, the sheep brains that were collected and processed by Charles Bagley, in the course of experimental physiological studies,⁶ were later used by Jerzy Rose for his classical descriptions of thalamic and cerebral cortical architecture.^{7,8} When John I. Johnson began neuroanatomical study, his first instruction from Wally Welker was to learn the organization of the sheep thalamus as described by Rose,⁸ and this particular exercise is still a most instructive and thorough introduction to the architecture of the mammalian thalamus.

In several reorganizations of neuroscience at Hopkins in the years after the Meyer era, Vernon Mountcastle, who studied with Meyer as a medical student, stayed at Hopkins his entire professional life, became chairman of the Hopkins physiology department, founder of the Hopkins neuroscience program, and was the first president of the

Society for Neuroscience. Seeing the value and the need for preservation of the priceless Meyer Collection, Mountcastle was instrumental in arranging its transfer to NMHM to join the Wisconsin and Michigan State Collections; themselves direct descendants, in outlook, methodology and organization, of the Meyer collection.

Unlike these descendant collections, the Meyer Phipps Collection was not designed to be a phylogenetic or zoological collection. This is reflected in the highly general naming of species and clades of the animals whose brains were collected originally for various experimental purposes. However, this defect as a systematic biological collection is offset to a degree by the fact that there exists a wealth of experimental articles containing physiological and behavioral data directly related to the neuroanatomy in the stained sections.

Contents of the collection

The largest component of the 267 specimens in the collection is a group of 197 mammal brains belonging to 56 species. Other holdings are 22 bird brains from 13 species, 34 brains of 11 species of reptiles, 6 brains of one amphibian species, 17 brains of 5 species of bony fish, and one shark brain (Table S4). The generally well-stained sections are embedded in celloidin: 169 specimens are stained for cells, 75 other specimens show myelinated fibers, and 15 have alternating fiber and cell-stained sections.

The Huber-Crosby and Crosby-Lauer Collections

History of the collections

The assembly of the Yakovlev, Wisconsin, and Michigan State collections at the NMHM attracted the attention of Sarah Winans Newman, chairperson of the Department of Anatomy and Cell Biology at the University of Michigan at Ann Arbor. The department had custody of the extensive assemblies of over 1,000 brain specimens collected by Elizabeth C. Crosby, working with G. Carl Huber until his passing, and with her profusion of students during her six decades of teaching and researching in human and comparative vertebrate neuroanatomy in Ann Arbor. Newman had concerns about the eventual fate of these extraordinary resources. With her support and the copious assistance of the curator of these materials, Beverly Shepard, this accumulation

of specimens of vertebrate brains was catalogued and transferred to the NMHM.

The collection is divided into two parts, an earlier Huber-Crosby and a later Crosby-Lauer Collection. Huber was the anatomical laboratory director who hired, collaborated with, and supported Crosby for more than two decades. Edward W. Lauer was a Crosby student who served many years as a faculty member in the department.

The Huber-Crosby Collection, along with that of C. U. Ariëns Kappers in Amsterdam (see the article by Hofman in this volume), formed the foundation of what became “The Comparative Anatomy of the Nervous System of Vertebrates, Including Man,”⁹ the three volume standard and encyclopedic reference for vertebrate neuroanatomy for over half a century, from the 1930s to the 1990s. The authors are listed as Ariëns Kappers, Huber and Crosby, but it is widely recognized that this listing should be in the reverse order, as is the case with the Huber-Crosby Collection. Crosby was deferential to Huber in memorial respect for his support and collaboration over the years, and listed his name before hers in both the book and in the naming of the collection. Crosby spent some time in Amsterdam with C. U. Ariëns Kappers, and was invited by him to undertake the translation,¹⁰ which soon became the revision and updating, of his earlier massive work, *Die vergleichende Anatomie des Nervensystems der Wirbeltiere und des Menschen*.^{11,12}

The later Crosby-Lauer Collection was largely assembled by Lauer, whom Crosby had encouraged to come to graduate school; he eventually became her fellow faculty member and continuing collaborator. This collection contains representatives of species not included in the Huber-Crosby Collection. A beautifully stained specimen of the brain of a greater panda, *Ailuropoda melanoleuca*, is listed in both collections; it was a resource used by Lauer in his study of the olfactory system in this species.¹³

Contents of the collections

The Huber-Crosby Collection, (Table S5A) along with that of C. U. Ariëns Kappers in Amsterdam, contains specimens of 15 cyclostomes, 5 sharks, 17 bony fish, 47 amphibians, 122 reptiles, 131 birds and 393 mammals, making a total of 730.

The Crosby-Lauer Collection (Table S5B) includes specimens of 2 cyclostomes, 5 bony fish, 5 amphibians, 7 reptiles, 14 birds, and 58 mammals.

Discussion

Problematic species identifications

For the listings of contents of all of the collections, we present the names as chosen, listed, and organized by the donors or curators of the collections at the time of accession. Many of the names are currently considered obsolete; however, obsolete names are often resurrected during long useful lives that can extend over decades and into centuries. More problematic are vague designations, such as “snake,” but it is dangerous to guess at the true identity of a specimen. It is conceivable that in the not too distant future, it will be possible to more securely identify specimens through DNA analysis of a bit of tissue. This is one importance of preserving actual tissues rather than only digital representations.

The importance of digitized renditions of collected specimens

However, there is no reason to demean the tremendous value of digital representations of collection specimens. The article by Jones, Stone, and Karten, in this volume, makes an excellent case for the value of modern imaging methods in making available information, resident in collection specimens, to widespread and varied audiences.

Wally Welker was a pioneer in using digital photography and the Internet in making available knowledge about the specimens he assembled, on the website <http://brainmuseum.org>. The brain museum images have become a standard source for educational and general publication purposes, when pictures of whole brains or sample sections are needed. We are making use of several of them in this paper to show some of the value of this material (Figs. 1–3). The article by Buchanan and Johnson in this volume illustrates how they can be used to advance comparative neuroscience.

According to Google statistics, The Human Brain Atlas at <https://www.msu.edu/user/brains/> is the most widely used online human brain atlas, and data retrieved from permission requests show that it is used at all educational levels on all continents, as well as in textbooks, research publications and the popular press. The atlas images are from the Yakovlev-Haleem Collection, an interesting use, for general information, of material originally assembled for pathological diagnoses and histories. The companion Sheep Brain Atlas at this site is in similar widespread use in veterinary education and for gen-

eral neuroscience laboratories where cheap sheep brains are the demonstration materials in common use. Images in the Sheep Brain Atlas are from the Welker Wisconsin and Johnson Michigan State collections. A unique Dolphin Brain Atlas at the same location has images of sagittal and horizontal sections from the Yakovlev-Haleem Collection; a series of images of coronal sections from the Welker Wisconsin Collection is in preparation for addition to the atlas.

These are but initial examples of the momentous impact brain collection material, in digital transcription, can have on all branches of human informational activities worldwide. Digital rendition of collection resources is a vital activity, and can be pursued as needed for specific purposes. Adoption of standard digitizing formats and methods can result, over long periods of time, in formidable databases assembled from individual digitizing projects.

Conflicts of interest

The authors declare no conflicts of interest.

Supporting Information

Additional supporting information may be found in the online version of this article:

Table S1: Detailed lists of specimens in the Yakovlev-Haleem Collection, useful for studies of vertebrate brain evolution. (A) List of normative control human specimens in sequence by age. (B) List of nonhuman vertebrate specimens.

Table S2: List of specimens in the Welker Wisconsin Collection, arranged alphabetically by common names of species. The specimen numbers designate the year of acquisition by the first two digits, followed by a hyphen and 1 to 3 digits denoting the order in which the specimens were acquired. Thus 63–361 designates the 361st specimen acquired in the Wisconsin laboratories in the year 1963. The stain listed as Heidenhain is the Sanides-Heidenhain-Woelcke modification of the Weigert method, introduced to the Wisconsin laboratories by Friedrich Sanides in the mid-1960s.

Table S3: List of specimens in the Johnson Michigan State Collection, arranged in approximate phylogenetic order, according to the Universal Species Identification System.⁵ The specimen numbers for

the Michigan State Collection carry the following information. The first two digits represent the year of acquisition. The third digit gives some information about the purpose of acquiring the specimen – 0 is a normal marsupial or monotreme specimen, 1 is an experimental specimen, 2 is a normal specimen, 3 is a cat, 4 is an opossum, and 5 is a raccoon. The last two digits represent the sequence during that year in which the specimen was acquired. Thus 65063 indicates a normal monotreme or marsupial specimen, the 63rd specimen acquired in 1965.

Table S4: List of specimens in the Meyer Phipps Johns Hopkins Collection.

Table S5: Lists of specimens in the Huber-Crosby and Crosby-Lauer collections.

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